

CPEN 416 / EECE 5710

Lecture 09: Grover's algorithm

Thursday 9 October 2025

Announcements

- Midterm available for pick up at my office
- Practice assignment 2 available after class
- Project details available ~~after class~~ *now*
- **No tutorial Monday** (Thanksgiving)
- Quiz 4 on Tuesday about *this week's* material

Last time

We modeled breaking a (binary) lock as a function:

$$f(\vec{x}) = \begin{cases} 1 & \vec{x} = \vec{s} \text{ the correct combo} \\ 0 & \text{otherwise} \end{cases}$$



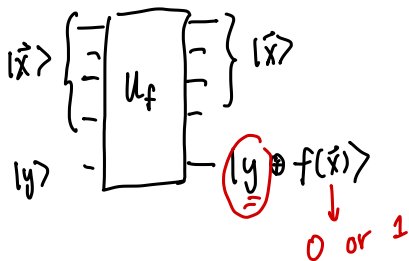
Trying combination $\mathbf{x}$ modeled as querying an *oracle* that evaluates $f(\mathbf{x})$. The number of queries is an algorithm's *query complexity*.

Image credit: Codebook node A.1

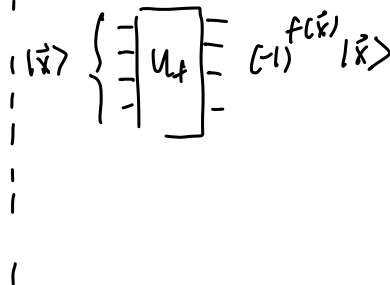
Last time

We expressed oracle queries as quantum circuits in two ways.

$$U_f |\vec{x}\rangle |y\rangle = |\vec{x}\rangle |y \oplus f(\vec{x})\rangle ; \quad U_f |\vec{x}\rangle = (-1)^{f(\vec{x})} |\vec{x}\rangle$$



$$|y\rangle = |1\rangle$$



Last time

We saw how phase kickback can be used to obtain information about multiple evaluations of $f(x)$ with a single query.

$$|x\rangle|-\rangle = \frac{1}{\sqrt{2}}|x\rangle|0\rangle - \frac{1}{\sqrt{2}}|x\rangle|1\rangle$$

$$\begin{aligned} U_f(|x\rangle|-\rangle) &= \frac{1}{\sqrt{2}}U_f(|x\rangle|0\rangle) - \frac{1}{\sqrt{2}}U_f(|x\rangle|1\rangle) \\ &= \frac{1}{\sqrt{2}}|x\rangle|f(x)\rangle - \frac{1}{\sqrt{2}}|x\rangle|1 \oplus f(x)\rangle \end{aligned}$$

kickback

$$f(x)=0 : \frac{1}{\sqrt{2}}|x\rangle|0\rangle - \frac{1}{\sqrt{2}}|x\rangle|1\rangle = |x\rangle|-\rangle$$

$$f(x)=1 : \frac{1}{\sqrt{2}}|x\rangle|1\rangle - \frac{1}{\sqrt{2}}|x\rangle|0\rangle = -|x\rangle|-\rangle$$

$$\Rightarrow U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle|-\rangle$$

$$\Rightarrow U_f\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)|-\rangle \rightarrow (-1)^{f(0)}\frac{1}{\sqrt{2}}|0\rangle|-\rangle + (-1)^{f(1)}\frac{1}{\sqrt{2}}|1\rangle|-\rangle$$

Last time

We saw how phase kickback can be used to obtain information about multiple evaluations of $f(x)$ with a single query.

$$\begin{aligned}
 \Rightarrow U_f \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) |-\rangle &\rightarrow (-1)^{f(0)} \frac{1}{\sqrt{2}} |0\rangle |-\rangle + (-1)^{f(1)} \frac{1}{\sqrt{2}} |1\rangle |-\rangle \\
 &= (-1)^{f(0)} \left[\frac{1}{\sqrt{2}} |0\rangle |-\rangle + (-1)^{f(0) \oplus f(1)} \frac{1}{\sqrt{2}} |1\rangle |-\rangle \right] \\
 &= (-1)^{f(0)} \left[\frac{1}{\sqrt{2}} (|0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle) |-\rangle \right]
 \end{aligned}$$

$f(0) = f(1)$ $f(0) \neq f(1)$

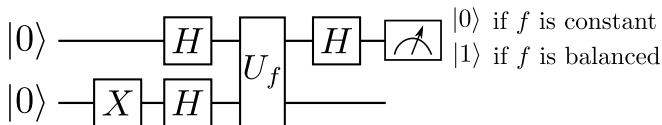
$|+\rangle |-\rangle$ $|-\rangle |-\rangle$

Last time

We applied Deutsch's quantum algorithm to determine if a function is *constant* or *balanced* using one oracle query

00 0 1
 01 0 1
 10 1 1
 11 1 1
 Bal Const.

Name	Action	Name	Action
f_1	$f_1(0) = 0$ $f_1(1) = 0$	f_2	$f_2(0) = 1$ $f_2(1) = 1$
f_3	$f_3(0) = 0$ $f_3(1) = 1$	f_4	$f_4(0) = 1$ $f_4(1) = 0$



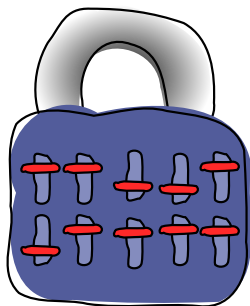
This is a quantum speedup; classical case requires 2 queries.

Learning outcomes

- Describe the strategy of amplitude amplification
- Derive the query complexity of Grover's search algorithm
- Implement Grover's search algorithm in PennyLane

Grover's quantum search algorithm

Let's break that lock!



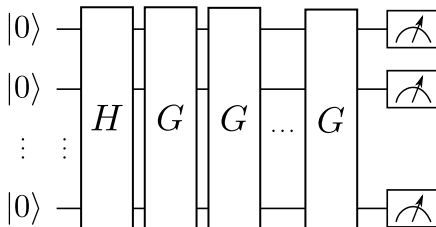
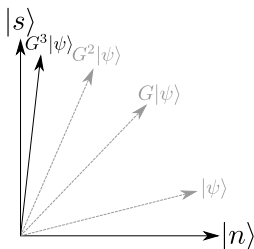
Classical: in the worst case, 2^n oracle queries

Quantum: $O(\sqrt{2^n})$ queries with Grover's algorithm

Image credit: Codebook node A.1

Grover's quantum search algorithm

Grover's search algorithm starts with a uniform superposition, then *amplifies* the amplitude of the state corresponding to the solution.



Grover's quantum search algorithm

In other words, we want to go from the uniform superposition

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} |\vec{x}\rangle$$

to something like this:

$$|\psi\rangle = (\text{big amplitude}) |\vec{s}\rangle + (\text{small amplitude}) \sum_{\vec{x} \neq \vec{s}} |\vec{x}\rangle$$

Amplitude amplification: geometric visualization

Subspace of
special $|s\rangle$



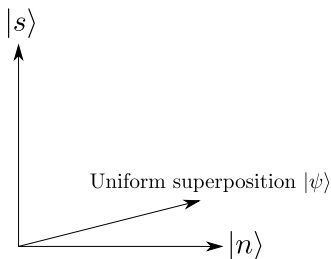
Subspace of
non-special $|x\rangle$



Partition the computational basis
states into two subspaces:

- 1 The special state $|s\rangle$
- 2 All the other states

Amplitude amplification: geometric visualization



$$|s\rangle = |\vec{s}\rangle$$

$$|n\rangle = \frac{1}{\sqrt{2^m-1}} \sum_{\vec{x} \neq \vec{s}} |\vec{x}\rangle$$

↑
m qubits

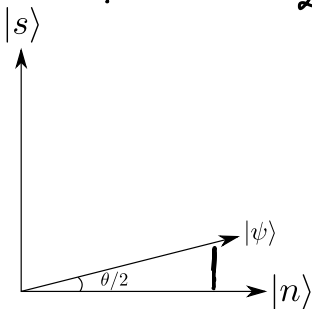
We can write the uniform superposition in terms of $|s\rangle$ and $|n\rangle$:

$$\begin{aligned} |\psi\rangle &= \frac{1}{\sqrt{2^m}} \sum_{\vec{x} \in \{0,1\}^m} |\vec{x}\rangle \\ &= \frac{1}{\sqrt{2^m}} \sum_{\vec{x} \neq \vec{s}} |\vec{x}\rangle + \frac{1}{\sqrt{2^m}} |\vec{s}\rangle \\ &= \frac{\sqrt{2^m-1}}{\sqrt{2^m}} |n\rangle + \frac{1}{\sqrt{2^m}} |s\rangle \end{aligned}$$

Amplitude amplification: geometric visualization

$$|\langle \psi | s \rangle|^2 = \frac{1}{2^n}$$

$$|\langle \psi | n \rangle|^2 = \frac{2^n - 1}{2^n}$$



Instead of working with these complicated coefficients

$$|\psi\rangle = \frac{1}{\sqrt{2^n}}|s\rangle + \frac{\sqrt{2^n - 1}}{\sqrt{2^n}}|n\rangle,$$

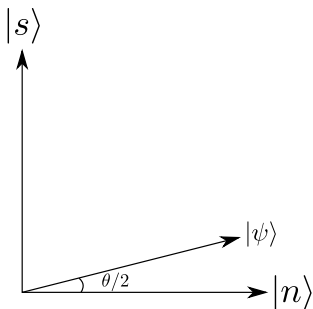
reexpress in terms of an angle θ :

$$|\psi\rangle = \sin\frac{\theta}{2}|s\rangle + \cos\left(\frac{\theta}{2}\right)|n\rangle$$

$$\downarrow$$

$$\cos\frac{\theta}{2} = \langle \psi | n \rangle = \frac{\sqrt{2^n - 1}}{\sqrt{2^n}}$$

Amplitude amplification: geometric visualization

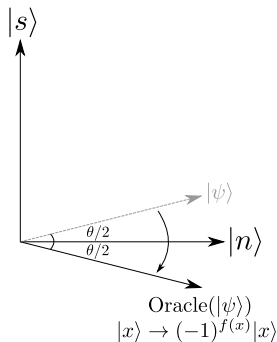


Apply operations to increase the amplitude of $|s\rangle$ while decreasing that of $|n\rangle$.

Two steps:

- 1 Apply the oracle O to 'pick out' the solution
- 2 Apply a 'diffusion operator' D to adjust the amplitudes.

Amplitude amplification: geometric visualization



$$O|\vec{x}\rangle = (-1)^{f(\vec{x})}|\vec{x}\rangle$$

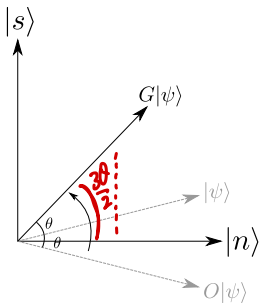
The oracle *flips* the amplitude of the special basis states in $|\psi\rangle$.

We can visualize this as a *reflection about the subspace* of non-special elements.

Exercise: how does this affect the uniform superposition state?

$$|\psi\rangle = \sin\frac{\theta}{2}|s\rangle + \cos\frac{\theta}{2}|n\rangle \longrightarrow -\sin\frac{\theta}{2}|s\rangle + \cos\frac{\theta}{2}|n\rangle$$

Amplitude amplification: geometric visualization

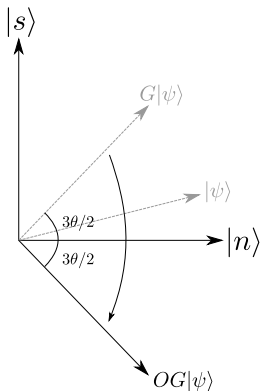


The diffusion operator is a bit less intuitive - it performs a *reflection about the uniform superposition state*.

Combined action $G = DO$ sends

$$G \left(\sin \left(\frac{\theta}{2} \right) |s\rangle + \cos \left(\frac{\theta}{2} \right) |n\rangle \right) = \sin \left(\frac{3\theta}{2} \right) |s\rangle + \cos \left(\frac{3\theta}{2} \right) |n\rangle$$

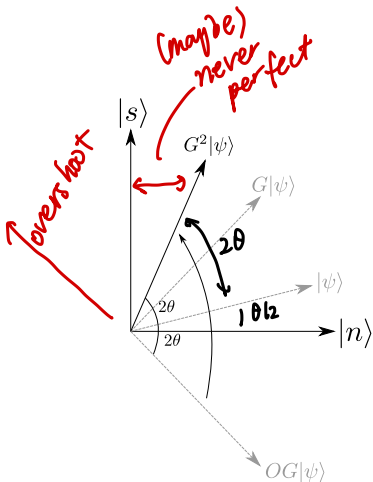
Grover's algorithm: geometric visualization



Grover's algorithm works by repeated application of $G = DO$ (called the Grover iteration).

Apply the oracle and reflect about the non-special elements.

Grover's algorithm: geometric visualization



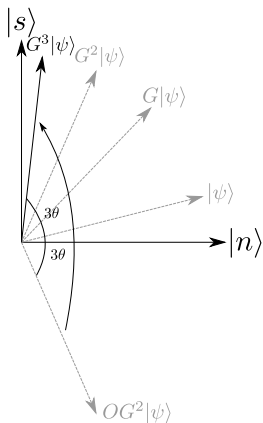
Apply the diffusion operator and reflect about the uniform superposition to boost the amplitude of the special state.

$$G^2|\psi\rangle = \sin\left(\frac{5\theta}{2}\right)|s\rangle + \cos\left(\frac{5\theta}{2}\right)|n\rangle$$

Grover's algorithm: geometric visualization

After k Grover iterations we will have the state

$$G^k|\psi\rangle = \sin\left(\frac{(2k+1)\theta}{2}\right)|s\rangle + \cos\left(\frac{(2k+1)\theta}{2}\right)|n\rangle$$



Exercise: what k maximizes the probability of measuring $|s\rangle$?

$$\sin\left(\frac{(2k+1)\theta}{2}\right) \rightarrow \text{max at } \frac{\pi}{2}$$

$$\sin\frac{\theta}{2} = \frac{1}{\sqrt{2^n}} \Rightarrow \theta = 2 \arcsin\left(\frac{1}{\sqrt{2^n}}\right)$$

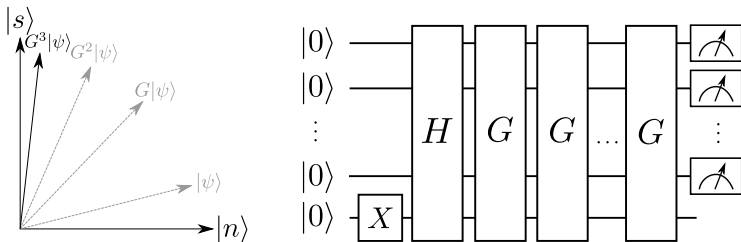
$$\text{optimal } k : \left\lfloor \frac{\pi}{4} \sqrt{2^n} \right\rfloor$$

quant. query complexity
is $O(\sqrt{2^n})$

Implementing Grover search

We will use the “type 1” oracle:

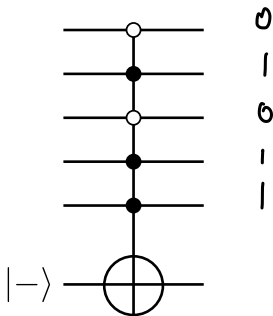
$$O|\mathbf{x}\rangle|y\rangle = |\mathbf{x}\rangle|y \oplus f(\mathbf{x})\rangle$$



What do circuits for the oracle and diffusion look like?

The oracle circuit

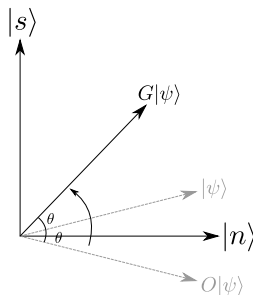
Exercise: show that a multicontrolled X gate, controlled on s , can be used as an oracle:



Try yourself: apply to
 $|x\rangle|-\rangle$ for $|x\rangle=|\vec{s}\rangle$
and $|x\rangle\neq|\vec{s}\rangle$

The diffusion circuit

The diffusion operator performs a reflection about the uniform superposition state.



The diffusion circuit

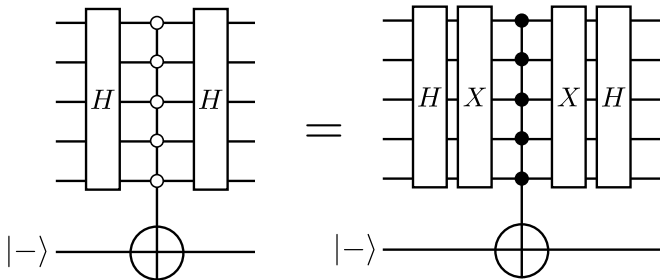
Exercise: Show that the unitary matrix given by

$$D = 2|\psi\rangle\langle\psi| - I$$

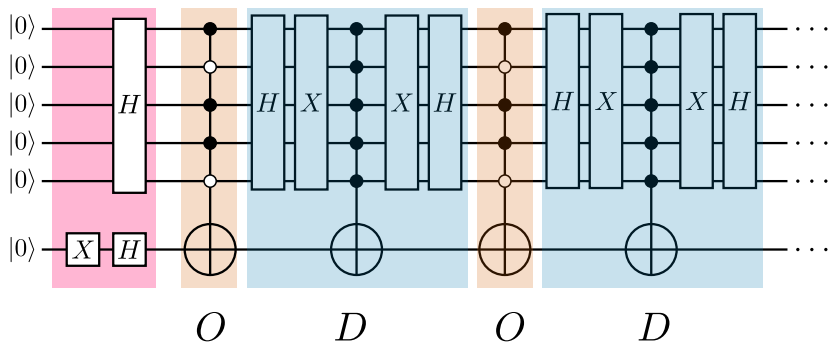
is equivalent to the diffusion operator.

Try yourself !

The diffusion circuit

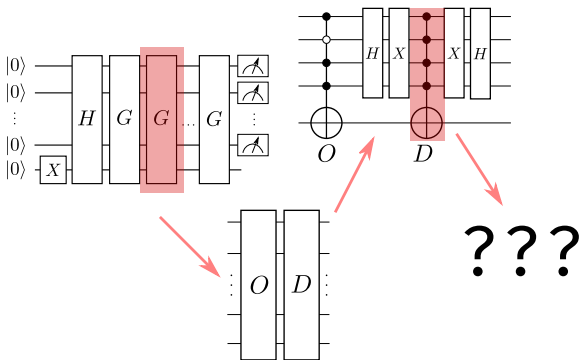


The full Grover circuit



The full Grover circuit

Clearly, each of the $O(\sqrt{2^n})$ queries requires some number of gates... how much does Grover *really* cost?



Next class: look inside the black box!

Next time

Content:

- Introduction to quantum compilation and resource estimation
- Quiz 4

Action items:

- ① Pick project group and paper
- ② Practice assignment 2

Recommended reading:

- For this class: Codebook module GA; Nielsen & Chuang 6.1
- For next class: Nielsen & Chuang 4.5

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